QUICK SORT:

Quick Sort is a comparison-based sorting algorithm that selects an element called a pivot and partitions the array so that smaller elements are placed on one side and larger elements on the other side.

ALGORITHM:

1.Select an element as the pivot.

2.Compare the other elements with the pivot.

3.Place smaller elements on the left side.

4.Place larger elements on the right side.

5.Put the pivot in its correct position.

6.Apply Quick Sort recursively to the left part.

7.Apply Quick Sort recursively to the right part.

8.Continue until the array is sorted.

Time Complexity:

Case	Complexity
Best	$O(n \log n)$
Average	$O(n \log n)$
Worst	$O(n^2)$

Space Complexity:

Space Complexity **= O(n)**

CONCLUSION:

Quick Sort is generally very fast and efficient in practice. Its average time complexity is **$O(n \log n)$**, but choosing a poor pivot can result in **$O(n^2)$** in the worst case.

OUTPUT:

```
Enter number of elements: 6
Enter elements:
9
5
7
8
2
4
Sorted array: 2 4 5 7 8 9

...Program finished with exit code 0
Press ENTER to exit console.
```